Book 1 — Lecture 1 — Interlude 1 — Exercise 2

Vector Subtraction Rule

Prompt. Work out the rule for vector subtraction.

Rule

Vector subtraction is defined componentwise:

$$\mathbf{A} - \mathbf{B} = (A_1 - B_1, \ldots, A_n - B_n).$$

Equivalent and often more useful form:

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}),$$

where $-\mathbf{B}$ is the additive inverse[1] of $\mathbf{B}$.

Geometric Interpretation

- If vectors share the same tail, $\mathbf{A} - \mathbf{B}$ is the vector from the tip[2] of $\mathbf{B}$ to the tip of $\mathbf{A}$.
- In 2D/3D drawings, this is equivalent to adding $\mathbf{A}$ to the reversed arrow $-\mathbf{B}$.

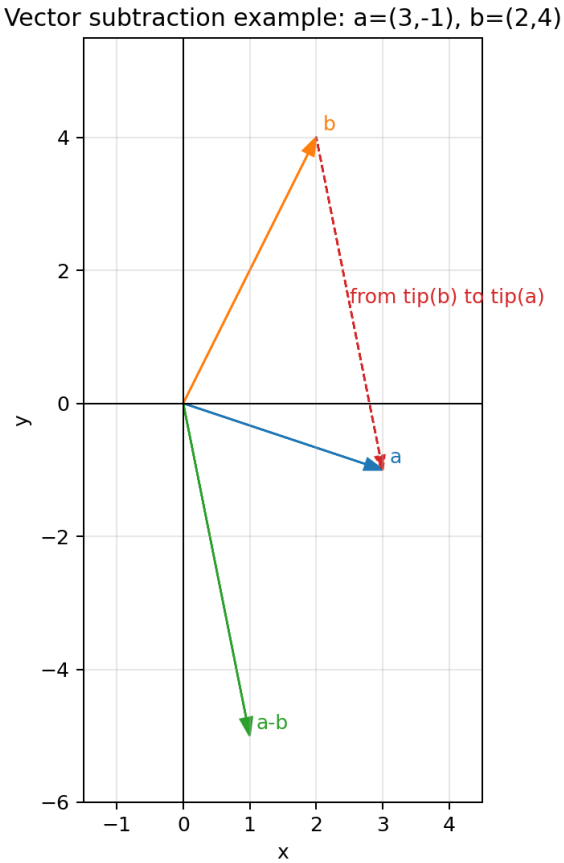
Quick Check in 2D

$$(3, -1) - (2, 4) = (3 - 2, -1 - 4) = (1, -5).$$

Result. Vector subtraction is componentwise and equivalent to adding the negative vector: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$.

Example. For $\mathbf{A} = (3, -1)$ and $\mathbf{B} = (2, 4)$, one gets $\mathbf{A} - \mathbf{B} = (1, -5)$.

[1] The additive inverse of $\mathbf{B}$ is the vector that sums with $\mathbf{B}$ to the zero vector.
[2] The *tip* is the arrowhead (terminal point) of a vector.

Example Drawing

The figure below visualizes $\mathbf{A} = (3, -1)$, $\mathbf{B} = (2, 4)$, and $\mathbf{A} - \mathbf{B} = (1, -5)$. The dashed red arrow shows the "tip of $\mathbf{B}$ to tip of $\mathbf{A}$" interpretation.